

CODE SPORT



In this month's column, we discuss some computer science interview questions.

As requested by a few of our readers, we decided to feature practice questions for computer science college students, for their upcoming on-campus and off-campus interviews. One small piece of advice to those preparing for interviews: Make sure that you are very clear about the fundamental concepts. Interviewers will not hold it against you if you don't know a specific algorithm. But not being able to answer some of the key fundamental concepts in different areas of computer science would definitely be an obstacle to clearing interviews. Also, make sure that you have updated your GitHub and SourceForge pages on which you might have shared some of your coding projects. Contributing to open source and having interesting, even if small, coding projects online in your GitHub pages would definitely be a big plus. In this month's column, we will discuss a medley of interview questions covering a wide range of topics such as algorithms, operating systems, computer architecture, machine learning and data science.

1. Given the advances in memory-driven computing, let us assume that you have been given an enterprise Class 16 core server with a huge 2TB RAM. You have been asked to design an intrusion detection system which can analyse the network logs of different systems and identify potential intrusion candidates. Note that you have to monitor close to 256 systems and analyse their network logs for potential issues. Let us also assume that you have been told which standard state-of-art approach (for intrusion detection) you would need to implement. Now the question is: Would you design the intrusion detection application as a scale up or a scale out application? Can you justify your choice? Would your choice change if,

instead of the enterprise Class 16 server with 2TB RAM, you are given multiple standard workstations with 32GB RAM.

2. You have been asked to build a supervised binary classification system for detecting spam emails. Basically, given an email, your system should be able to classify it as spam or non-spam. You are given a training data set containing 400,000 rows and 10,000 columns (features). The 10,000 features include both numerical variables and categorical variables. Now, you need to create a model using this training data set. Consider the two options given in Question 1 above, namely a server with 2TB RAM or a simple workstation with 32GB RAM. For each of these two options, can you explain how your approach would change in training a model? (*Clue:* The workstation is memory constrained, hence you will need to figure out ways to reduce your data set. Think about dimensionality reduction, sampling, etc.) Now, assume that you are able to successfully create two different models — one for the enterprise server and another for the workstation. Which of these two models would do better on a held-out test data set? Is it always the case that the model you built for the enterprise class server with huge RAM would have a much lower test error than the model you built for the workstation? Explain the rationale behind your answer. Here is a tougher bonus question: For this specific computer system which has a huge physical memory, do you really need paging mechanisms in the operating system? If you are told that all your computer systems would have such huge physical memory, would you be able to completely eliminate the virtual memory